

Procedural Sedation with Laryngospasm



Section I: Scenario Demographics

Scenario Title:	Procedural Sedation with Laryngospasm		
Date of Development:	21/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a rare complication of a common procedure.
CRM Objectives:	Exercise leadership by actively distributing the workload through an unexpected crisis.
Medical Objectives:	1) Choose an appropriate agent for procedural sedation of a healthy child. 2) Recognize laryngospasm and initiate BVM ventilation for treatment. 3) Administer paralytic and intubate for persistent hypoxia in laryngospasm. 4) Administer atropine for succinylcholine-induced bradycardia.

Case Summary: Brief Summary of Case Progression and Major Events

The emergency team is preparing to perform a conscious sedation on a 7-year-old boy to facilitate the reduction of a fracture of the radius and ulna. They will be expected to do an airway assessment and pick an appropriate agent for sedation. In the middle of sedation, the patient's oxygen saturation will suddenly drop and the patient will stop breathing. The team will be unable to bag the patient until they ask for either deeper sedation or a paralytic. If they administer succinylcholine, the patient will become bradycardic and require atropine.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- Hobaika AB, Lorentz MN. [Laryngospasm]. Rev Bras Anesthesiol. 2009 Jul-Aug;59(4):487-95. Review. Portuguese. PubMed PMID: 19669024
- Larson CP Jr. Laryngospasm—the best treatment. Anesthesiology. 1998 Nov;89(5):1293-4. PubMed PMID: 9822036
- <http://lifeinthefastlane.com/ccclaryngospasm/>
- Al-Metwalli RR, Mowafi HA, Ismail SA: Gentle chest compression relieves extubation laryngospasm in children. J Anesth 2010; 24:854–7. PMID: 20976504
- <http://first10em.com/2015/03/04/laryngospasm/>



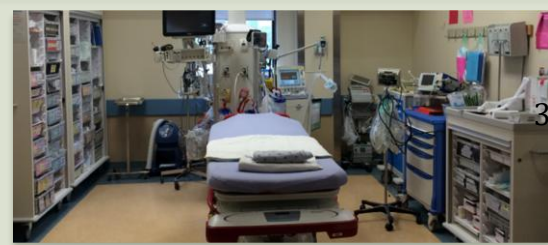
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Pediatric Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☐ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
Mother	To help with airway history		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: Casting material	
☐ Intraosseous Set-up	☐ LMA	☒ Other: Peds NRB PEEP valve	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	10 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

Cayden McFadden is a 7-year-old boy who has a fracture through the distal radius and ulna that requires reduction. The emergency physician treating him has just asked you to come provide procedural sedation for the reduction.

B. Patient Profile and History

Patient Name: Cayden McFadden	Age: 7	Weight: 28 kg
Gender: ☒ M ☐ F	Code Status: Full.	
Chief Complaint: Distal radius/ulna #		
History of Presenting Illness: Fell playing hockey. Requires sedation for reduction of fracture.		
Past Medical History:	None. No family hx anesthetic complications	Medications: None.

Allergies: None.

Social History: Lives with parents.

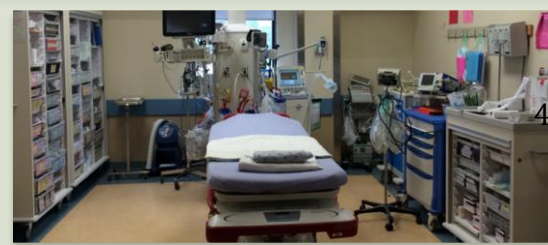
Family History: Non-contributory.

Review of Systems:	CNS:	Nil.
	HEENT:	Nil.
	CVS:	Nil.
	RESP:	Nil.
	GI:	Nil.
	GU:	Nil.
	MSK:	Pain to R wrist since injury.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 110/min	BP: 100/60	RR: 20/min
O ₂ SAT: 99% RA		
Rhythm: Sinus	T: 36.1°C	Glucose: 6.2 mmol/L
GCS: 15 (E4 V5 M6)		
General Status: Well.		
CNS:	Alert, oriented, appropriate.	
HEENT:	Nil.	
CVS:	Normal HS. No murmur.	
RESP:	GAEB. No adventitious. Normal mouth opening. Malampatti 1.	
ABDO:	Soft, NT.	
GU:	Nil.	
MSK:	Good neck ROM.	SKIN: Nil.

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: HR: 110/min BP: 100/60 RR: 20/min O ₂ SAT: 99% RA T: 36.1°C	Alert, appropriate. Sitting in bed with mom at bedside. Splint on R forearm.	<u>Learner Actions</u> - ☐ Ensure iv access - ☐ Preoxygenate patient - ☐ Check respiratory equipment - ☐ Perform airway assessment - ☐ Prepare procedural sedation meds - ☐ Full hx/px relevant to sedation	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Preoxygenate → O ₂ SAT 100% <u>Triggers</u> <i>For progression to next state</i> - Start sedation → 2. Sedation Start
2. Sedation Start Unchanged.	Patient slowly becomes less awake as sedation has appropriate effect	<u>Learner Actions</u> - ☐ Administer sedation (Ketamine and/or propofol) - ☐ Assess readiness for procedure - ☐ Ask mom to leave room - ☐ Allow other Emerg MD to start reduction	<u>Modifiers</u> - Start procedure without assessing LOC → patient says "ow" and "stop that" <u>Triggers</u> - 1 minute into manipulation of wrist → 3. Laryngospasm
3. Laryngospasm O ₂ SAT → 65% over 60 sec. HR → 130 RR → 0 BP → 130/70	Patient not responsive.	<u>Learner Actions</u> - ☐ BVM ventilation with PEEP valve (difficult to bag with poor chest rise) - ☐ Tell ED MD to stop reduction - ☐ Apply pressure to Larson's point with jaw thrust - ☐ Administer deeper sedation (propofol) - ☐ Administer paralytic - ☐ ± Intubate after paralytic - ☐ Gentle compression of chest	<u>Modifiers</u> - 1 min of BVM → O ₂ SAT to 75% <u>Triggers</u> - Succinylcholine given → 4. Bradycardia - 2 min of laryngospasm → 4. Bradycardia
4. Bradycardia O ₂ SAT → 95% over 1 min of bagging HR → 35 over 1 min		<u>Learner Actions</u> - ☐ Give atropine 0.02mg/kg iv (0.56mg) - ☐ Continue bagging patient if not yet intubated	<u>Modifiers</u> <u>Triggers</u> - Atropine given → 5. Resolution - 12 minutes → 5. Resolution
5. Resolution HR → to 110 over 1 min		<u>Learner Actions</u> - ☐ Disclose incident to mother	END CASE when heart rate normalizes

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work required.

Images (ECGs, CXRs, etc.)

No imaging required.

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) Explain the progression of the case today. In summary, what happened? 2) What is your approach to laryngospasm? 3) What is Larson's point? 4) What are some other complications of procedural sedation with ketamine? 5) How did it feel as a team to manage this extremely unexpected complication? Do you feel you remained calm? 6) How did the leader address the critical nature of this complication? As a team, did you feel that you had clear directions?	
Key Moments	
Airway assessment and preparation	
Recognition of Laryngospasm	
Treatment of medication-induced bradycardia	